Machine Learning
Homework 2

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1. Let $\{(x^{(i)}, y^{(i)})\}_{i=1}^{N}$ be N training samples, with $x^{(i)} \in \mathbb{R}^d$ and $y^{(i)} \in \{-1, +1\}$. Here N and d are positive integers. Consider the ℓ_2 soft margin SVM:

$$\begin{aligned} &\min_{w,b,\xi} \frac{1}{2}\|w\|_2^2 + \frac{C}{2}\sum_{i=1}^{N}\xi_i^2 \\ &\text{s.t.} \quad y^{(i)}(w^T x^{(i)} + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1, \ldots, N. \end{aligned} \tag{1}$$

The variables are $w \in \mathbb{R}^d$, $b \in \mathbb{R}$, and $\xi = (\xi_1, \ldots, \xi_N)^T \in \mathbb{R}^N$; $C > 0$ is given and $\|\cdot\|_2$ is the Euclidean norm.

(a) For fixed w and b, find the minimizing slack ξ_i for each sample. Substitute your result into the objective to obtain an unconstrained problem in w and b.

(b) The constraints $\xi_i \geq 0$ can be omitted without changing the optimal value. For this equivalent formulation, write the Lagrangian $\mathcal{L}(w, b, \xi, \alpha)$, where $\alpha = (\alpha_1, \ldots, \alpha_N)^T \in \mathbb{R}^N$ collects the nonnegative margin-constraint multipliers α_i.

(c) Set the derivatives of $\mathcal{L}$ with respect to w, b, and each ξ_i equal to zero, and give the resulting relations.

(d) Derive the dual problem, specifying its objective and all constraints on α.

Ans:

(a) For fixed w and b, define the signed margin $m_i = y^{(i)}(w^T x^{(i)} + b)$ for sample i. The constraints require $\xi_i \geq \max\{0, 1 - m_i\}$. Since $C > 0$ and ξ_i^2 increases on $[0, \infty)$, the minimizing slack is

$$\xi_i = \max\{0, 1 - m_i\} = \max\{0, 1 - y^{(i)}(w^T x^{(i)} + b)\}.$$

Substitution gives the equivalent unconstrained problem

$$\min_{\substack{w \in \mathbb{R}^d \\ b \in \mathbb{R}}} \frac{1}{2}\|w\|_2^2 + \frac{C}{2}\sum_{i=1}^{N}\big(\max\{0, 1 - y^{(i)}(w^T x^{(i)} + b)\}\big)^2.$$

The summand is the squared hinge loss.

(b) Without $\xi_i \geq 0$, the slack still minimizes ξ_i^2 subject to $\xi_i \geq 1 - m_i$. Its minimizer is again $\max\{0, 1 - m_i\}$. Thus omitting the nonnegativity constraints leaves the optimal value unchanged. Writing the remaining inequalities as $1 - \xi_i - y^{(i)}(w^T x^{(i)} + b) \leq 0$, we obtain

$$\mathcal{L}(w, b, \xi, \alpha) = \frac{1}{2} w^T w + \frac{C}{2} \sum_{i=1}^{N} \xi_i^2 + \sum_{i=1}^{N} \alpha_i [1 - \xi_i - y^{(i)}(w^T x^{(i)} + b)].$$

Here α is the multiplier vector defined in the question, with $\alpha_i \geq 0$ for $i = 1, \ldots, N$.

(c) For fixed α, stationarity gives

$$\begin{aligned} \nabla_w \mathcal{L} = w - \sum_{i=1}^{N} \alpha_i y^{(i)} x^{(i)} = 0 \quad &\Longrightarrow \quad w = \sum_{i=1}^{N} \alpha_i y^{(i)} x^{(i)}, \\ \frac{\partial \mathcal{L}}{\partial b} = -\sum_{i=1}^{N} \alpha_i y^{(i)} = 0 \quad &\Longrightarrow \quad \sum_{i=1}^{N} \alpha_i y^{(i)} = 0, \\ \frac{\partial \mathcal{L}}{\partial \xi_i} = C\xi_i - \alpha_i = 0 \quad &\Longrightarrow \quad \xi_i = \frac{\alpha_i}{C}. \end{aligned}$$

(d) Define the dual function by $g(\alpha) = \inf_{w,b,\xi} \mathcal{L}(w, b, \xi, \alpha)$, with $w \in \mathbb{R}^d$, $b \in \mathbb{R}$, and $\xi \in \mathbb{R}^N$ in this infimum. If $\sum_{i=1}^{N} \alpha_i y^{(i)} \neq 0$, the term linear in b is unbounded below, so $g(\alpha) = -\infty$. Otherwise, substitution of the minimizing w and ξ from part (c) gives

$$g(\alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \left\| \sum_{i=1}^{N} \alpha_i y^{(i)} x^{(i)} \right\|_2^2 - \frac{1}{2C} \sum_{i=1}^{N} \alpha_i^2.$$

Hence the dual problem is

$$\begin{aligned} \max_{\alpha \in \mathbb{R}^N} \quad & \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y^{(i)} y^{(j)} (x^{(i)})^T x^{(j)} - \frac{1}{2C} \sum_{i=1}^{N} \alpha_i^2 \\ \text{s.t.} \quad & \alpha_i \geq 0 \ (i = 1, \ldots, N), \qquad \sum_{i=1}^{N} \alpha_i y^{(i)} = 0. \end{aligned}$$

There is no upper bound $\alpha_i \leq C$. The primal problem is convex and strictly feasible (choose $w = 0$, $b = 0$, $\xi_i = 2$ for every i), so strong duality holds.

2. For a positive integer n and variable $x = (x_1, \ldots, x_n)^T \in \mathbb{R}^n$, consider

$$\begin{aligned} \min_x \; & -\sum_{i=1}^n \log(a_i + x_i) + \sum_{i=1}^n \beta_i x_i \\ \text{s.t.} \quad & x_i \geq 0 \; (i = 1, \ldots, n), \qquad \sum_{i=1}^n x_i = B. \end{aligned} \tag{2}$$

The parameter vectors $a = (a_1, \ldots, a_n)^T$ and $\beta = (\beta_1, \ldots, \beta_n)^T$ are given, with $a_i > 0$, $\beta_i \in \mathbb{R}$, and $B > 0$. The logarithm is natural.

(a) Write the Lagrangian and KKT conditions. Define all multipliers.

(b) Let x^* be the optimal solution and μ^* the multiplier of $\sum_{i=1}^n x_i = B$ at the optimum. Express x_i^* in terms of μ^* and give an equation determining μ^*. Then find x^* for $n = 3$, $a = (1, 2, 1)^T$, $\beta = (0, 0, 1)^T$, and $B = 2$.

Ans:

(a) Let $\lambda = (\lambda_1, \ldots, \lambda_n)^T$ with $\lambda_i \geq 0$ be the multipliers for $-x_i \leq 0$, and let $\mu \in \mathbb{R}$ be the multiplier for $\sum_{i=1}^n x_i - B = 0$. Define

$$\mathcal{L}(x, \lambda, \mu) = -\sum_{i=1}^n \log(a_i + x_i) + \sum_{i=1}^n \beta_i x_i - \sum_{i=1}^n \lambda_i x_i + \mu \left(\sum_{i=1}^n x_i - B \right).$$

A superscript $*$ denotes a value at the optimum: x^* is the primal solution and (λ^*, μ^*) are its associated multipliers. The KKT conditions are

$$\begin{aligned} & x_i^* \geq 0, \qquad \sum_{i=1}^n x_i^* = B && \text{(primal feasibility)}, \\ & \lambda_i^* \geq 0 && \text{(dual feasibility)}, \\ & \lambda_i^* x_i^* = 0 && \text{(complementary slackness)}, \\ & -\frac{1}{a_i + x_i^*} + \beta_i - \lambda_i^* + \mu^* = 0 && \text{(stationarity)}. \end{aligned}$$

Every condition indexed by i holds for $i = 1, \ldots, n$; the equality multiplier μ^* is unrestricted in sign. The objective has positive diagonal second derivatives $1/(a_i + x_i)^2$ and zero mixed derivatives, so it is strictly convex. Since $x_i = B/n > 0$ is strictly feasible, the KKT conditions characterize the unique primal optimum.

(b) Stationarity implies, for each i,

$$\beta_i + \mu^* = \frac{1}{a_i + x_i^*} + \lambda_i^* > 0.$$

Complementary slackness and dual feasibility give

$$x_i^* > 0 \quad \Longrightarrow \quad \lambda_i^* = 0, \quad x_i^* = \frac{1}{\beta_i + \mu^*} - a_i > 0,$$
$$x_i^* = 0 \quad \Longrightarrow \quad \lambda_i^* = \beta_i + \mu^* - \frac{1}{a_i} \geq 0.$$

Thus

$$\boxed{x_i^* = \max\left\{0, \frac{1}{\beta_i + \mu^*} - a_i\right\}, \quad i = 1, \ldots, n.}$$

The equality constraint determines μ^* through

$$\sum_{i=1}^{n} \max\left\{0, \frac{1}{\beta_i + \mu^*} - a_i\right\} = B, \qquad \mu^* > -\min_{1 \leq i \leq n} \beta_i.$$

On this domain, the sum decreases continuously from $+\infty$ to 0, strictly whenever positive. Therefore it equals $B > 0$ at exactly one value of μ.

For the stated example, the domain is $\mu^* > 0$, so

$$x_3^* = \max\{0, 1/(1 + \mu^*) - 1\} = 0.$$

Assuming the first two components are positive gives

$$\left(\frac{1}{\mu^*} - 1\right) + \left(\frac{1}{\mu^*} - 2\right) = 2, \qquad \mu^* = \frac{2}{5}.$$

This value makes both components positive, verifying the assumption. The solution and multipliers are

$$\boxed{x^* = (3/2,\, 1/2,\, 0)^T, \qquad \lambda^* = (0,\, 0,\, 2/5)^T, \qquad \mu^* = 2/5.}$$

Indeed, x^* is nonnegative and sums to 2, λ^* is nonnegative, and $\lambda_i^* x_i^* = 0$. The three stationarity equations evaluate to $-2/5+2/5 = 0$, $-2/5+2/5 = 0$, and $-1 + 1 - 2/5 + 2/5 = 0$, respectively.

3. For input vectors $x, z \in \mathbb{R}^2$, consider three SVM kernels: linear $K_{\text{lin}}(x, z) = x^T z$, quadratic $K_{\text{quad}}(x, z) = (x^T z + 1)^2$, and Gaussian $K_{\text{RBF}}(x, z) = e^{-\|x-z\|_2^2/(2\sigma^2)}$. Here $\|\cdot\|_2$ is the Euclidean norm and $\sigma > 0$ is the bandwidth. Give brief explanations; no derivations are required.

   (a) Which kernel is the simplest choice for linearly separable classes? Why?

   (b) Can a linear kernel represent a circular boundary? Suggest a suitable nonlinear candidate and explain. Several choices may be valid.

   (c) A Gaussian-kernel SVM produces many small bends around training samples. To make each sample's influence less local, should σ be increased or decreased? Keep other settings fixed and explain.

Ans:

(a) The linear kernel K_{lin} is the simplest suitable choice. Its nondegenerate decision boundary in $\mathbb{R}^2$ is a straight line, which is sufficient for the stated class arrangement. A nonlinear kernel is not required by this geometry.

(b) A linear kernel cannot represent a closed circular boundary: its nondegenerate boundary is a straight line. The quadratic kernel K_{quad} is a suitable choice because it can represent quadratic boundaries, including circles. The Gaussian kernel K_{RBF} is also a reasonable choice for this nonlinear pattern. The boundary's shape alone does not uniquely determine which kernel was used.

(c) First try increasing σ. A larger Gaussian bandwidth makes the kernel similarity decay more slowly with distance, so each sample influences a broader neighborhood. This encourages less localized behavior and may reduce small bends around individual samples. It does not guarantee a smoother fitted boundary for every dataset; the data and other training settings also matter.